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第二章　函数的导数(Derivative)

2.1　导数的定义(The Definition of Derivative)

Multiple Choice Questions

$\frac{2-1}{3}$

1. Which of the following values is the average rate of change of $f(x)=\sqrt{x+2}$ over the interval $[-1,2]$?

C

(A) -4　(B) $-\frac{1}{3}$　(C) $\frac{1}{3}$　(D) 3

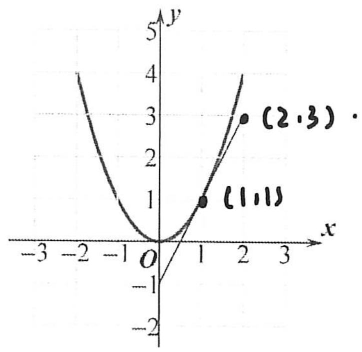
$4-3x-3h\quad -4+3x$

C

2. Let $f(x)=4-3x$. Which of the following is equal to $f'(0)$?

(A) 4　(B) -4　(C) -3　(D) 4

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1)=($ D $)$.



(A) 0　(B) $\frac{1}{2}$　(C) 1　(D) 2

$-8k+2+\cancel{8}k+2=9$

$-7k=5$

$k=-\frac{5}{7}$

4. Let $f(x)=kx^3-2x$, $(k\neq 0)$. The average rate of $f(x)$ over the interval $[-2,-1]$ is 3, find k.

D

(A) $\frac{5}{7}$　(B) $\frac{1}{7}$　(C) $-\frac{1}{7}$　(D) $-\frac{5}{7}$

C

5. **Caculator** Let $y=\tan e^x-x^2+1$. Find $\frac{dy}{dx}\Big|_{x=0.5}=$

(A) -12.056　(B) 0　(C) 271.066　(D) 273.066

6. **Caculator** The graph of $y=\sqrt{x^2+1}-3x$ crosses the x-axis at one point in the interval $[0,1]$. What is the slope of the graph at this point?

C

(A) 0.354　(B) -0.001　(C) -2.666　(D) 0

7. The population of insects at time t days is modeled by the function P with first derivative $P'(t)=0.1t^2+10t+120$. Which of the following statements is true?

(A) At time $t=10$, the population of insects is 230.

B

(B) At time $t=10$, the population of insects is increasing at a rate of 230 insects per day.

(C) At time $t=10$, the population of insects is decreasing at a rate of 230 insects per day.

(D) At time $t=10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

A

8. If the line tangent to the graph of a function g at the point (1,4) passes through the point $(-2,-5)$, what is the slope of this line?

(A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

Free Response Questions

9. Let $f(x)=x^2-2x$

(a) Find the average rate of change of f over $[1,3]$.

(b) Find $\frac{f(2+h)-f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x=2$.

a) $k=\frac{9-6-1+2}{2}=2$

b) 求 $\frac{2^2+4h+h^2-0-4}{h}=4+h$

c) $f'(2)=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=4+0=4$

10. Use the definition of derivative to find the derivative of $f(x)=x^3$, then find $f'(2)$.

$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\frac{(x+h)^3-x^3}{h}$

$=\frac{2hx^2+h^2x+hx^2+2h^2x+h^3}{h}$

$=2x^2+hx+x^2+2h+h^2$

$=3x^2$

$f'(2)=3\times 4=12$